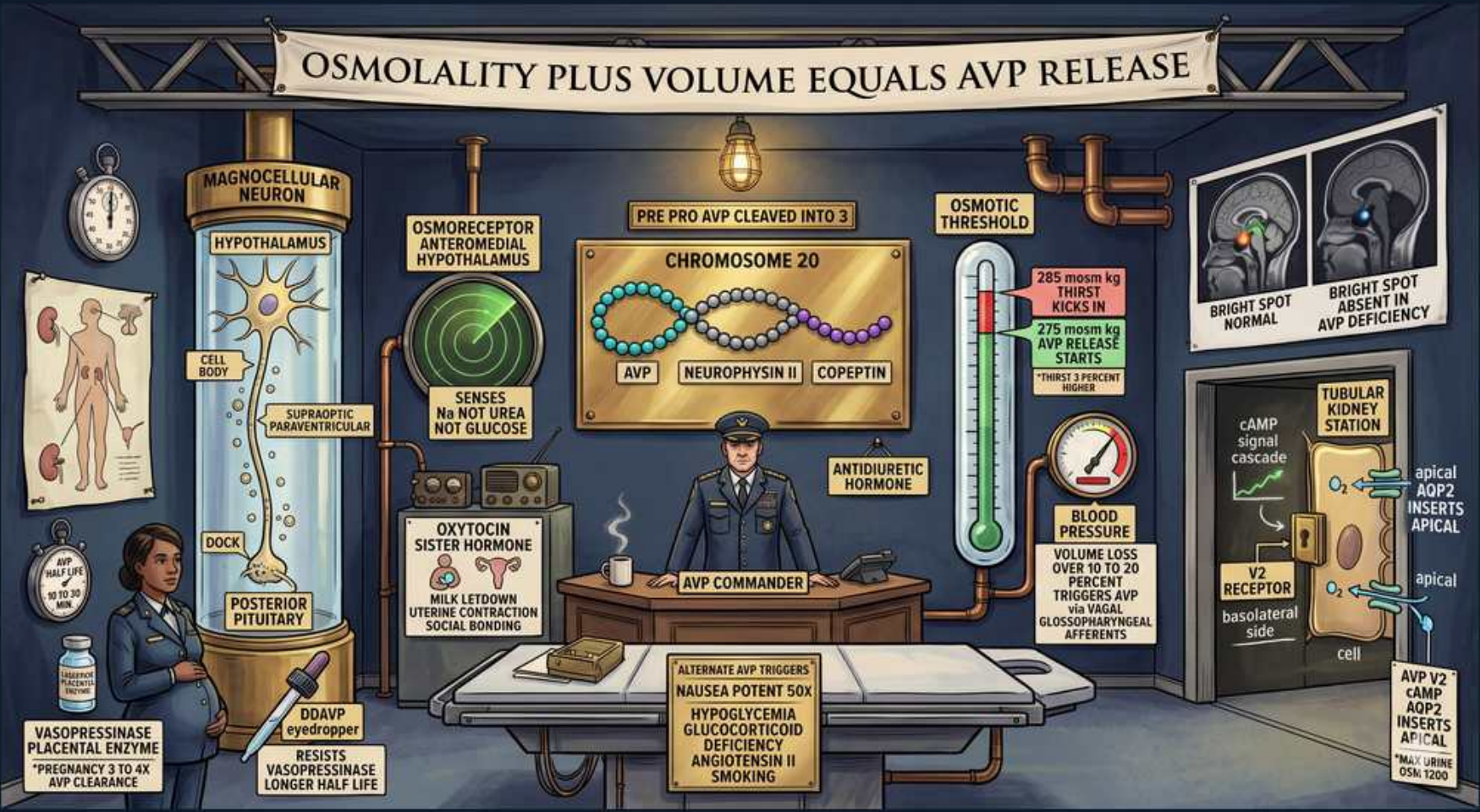


Posterior pituitary — AVP/ADH release physiology



1. Magnocellular neuron

WHAT YOU SEE

Labeled neuron with cell body in hypothalamus

WHAT IT ENCODES

AVP is synthesized in magnocellular neurons of the hypothalamic supraoptic and paraventricular nuclei. The hormone travels down axons to the posterior pituitary for storage and release. This is true neurosecretion, not endocrine gland secretion.

BOARD TRAP

Anterior pituitary makes ACTH/TSH/GH — but ADH/oxytocin are made in the hypothalamus and only STORED in the posterior pituitary.

2. Posterior pituitary storage

WHAT YOU SEE

Posterior pituitary at base of neuron

WHAT IT ENCODES

The posterior pituitary (neurohypophysis) is a storage/release site, with no hormone synthesis of its own. AVP and oxytocin are released here into the systemic circulation. Pituitary stalk section therefore impairs release.

BOARD TRAP

A stalk lesion can cause transient central DI; distinguish from anterior hypopituitarism which causes ACTH/TSH/gonadotropin loss.

3. Chromosome 20 — AVP gene

WHAT YOU SEE

DNA helix labeled chromosome 20, AVP-neurophysin II-copeptin

WHAT IT ENCODES

The preprohormone is cleaved into AVP, neurophysin II, and copeptin in a 1:1:1 ratio. Copeptin is the stable C-terminal fragment used as a surrogate marker for AVP secretion. Familial central DI maps to AVP-neurophysin II gene mutations.

BOARD TRAP

Copeptin (not AVP itself) is measured clinically because AVP is unstable and platelet-bound; high copeptin distinguishes nephrogenic DI from central DI.

4. Osmoreceptor

WHAT YOU SEE

Anterolateral hypothalamus osmoreceptor station

WHAT IT ENCODES

Anterior hypothalamic osmoreceptors sense plasma osmolality and drive AVP release. They respond to effective osmoles like sodium, not to urea or glucose which cross cell membranes freely. A 1-2% rise in osmolality triggers secretion.

BOARD TRAP

Urea and glucose are ineffective osmoles for osmoreceptors — uremia or hyperglycemia raise measured osmolality without driving AVP through this pathway.

5. Osmotic threshold

WHAT YOU SEE

Thermometer marked 280 and 285 mosm/kg

WHAT IT ENCODES

AVP release begins around 280-285 mosm/kg; thirst is activated at a slightly higher threshold near 290-295 mosm/kg. This staggering keeps plasma osmolality tightly regulated between roughly 280-295. Resetting of this threshold occurs in pregnancy and SIADH.

BOARD TRAP

In SIADH the osmostat is reset/inappropriately active — AVP is secreted despite low osmolality, the opposite of the normal threshold logic.

6. Thirst threshold ~290

WHAT YOU SEE

Marker 290 mosm/kg thirst kicks in

WHAT IT ENCODES

Thirst is the final defense against hyperosmolality and is activated above the AVP threshold. An intact thirst mechanism normally prevents severe hypernatremia even in DI. Loss of thirst (adipsia) plus DI is dangerous.

BOARD TRAP

A DI patient with intact thirst maintains near-normal sodium by drinking; hypernatremia in DI implies impaired access to water or adipsic DI.

7. Volume/baroreceptor stimulus

WHAT YOU SEE

Blood pressure gauge, volume loss 10-20 percent

WHAT IT ENCODES

A large fall in blood volume or pressure (about 10-20%) is a potent non-osmotic stimulus for AVP via baroreceptor afferents. This volume stimulus overrides osmotic control and can cause hyponatremia in hypovolemia. It explains ADH excess in heart failure and cirrhosis.

BOARD TRAP

Hypovolemic hyponatremia: ADH is appropriately high from baroreceptor drive — this is NOT SIADH (which requires euvolemia).

8. V2 receptor — kidney

WHAT YOU SEE

Tubular kidney station, V2 receptor basolateral

WHAT IT ENCODES

AVP acts on V2 receptors on the basolateral collecting duct membrane, raising cAMP and inserting aquaporin-2 into the apical membrane. This drives free water reabsorption and concentrated urine. Nephrogenic DI results from V2 receptor (X-linked) or AQP2 mutations.

BOARD TRAP

V2 receptor antagonists (vaptans) treat SIADH; congenital nephrogenic DI is most often X-linked V2 receptor mutation, not AQP2.

9. Aquaporin-2 insertion

WHAT YOU SEE

Apical AQP2 water channels inserted

WHAT IT ENCODES

cAMP triggers trafficking of AQP2 water channels to the apical membrane of principal cells. Water then moves down the medullary osmotic gradient, concentrating urine. Lithium impairs this cascade, causing acquired nephrogenic DI.

BOARD TRAP

Lithium and hypercalcemia/hypokalemia cause acquired nephrogenic DI by disrupting AQP2 — copeptin/AVP is HIGH, distinguishing from central DI.

10. cAMP signal cascade

WHAT YOU SEE

cAMP signaling labeled at kidney station

WHAT IT ENCODES

V2 receptor couples to Gs protein, activating adenylate cyclase and raising cAMP. cAMP-dependent protein kinase A phosphorylates AQP2 for membrane insertion. This is the molecular basis of urine concentration.

BOARD TRAP

Gs/cAMP cascade — congenital nephrogenic DI mutations in V2 (X-linked) block this step despite normal AVP levels.

11. Bright spot on MRI

WHAT YOU SEE

Two brain MRIs, bright posterior pituitary spot vs absent

WHAT IT ENCODES

The normal posterior pituitary shows a bright T1 spot from stored AVP-neurophysin. The bright spot is absent in central AVP deficiency. This is a useful but not definitive imaging clue.

BOARD TRAP

Absence of the bright spot supports central DI but can occur with age/normal variants — use copeptin testing for definitive diagnosis.

12. Oxytocin — sister hormone

WHAT YOU SEE

Oxytocin labeled, milk letdown and uterine contraction

WHAT IT ENCODES

Oxytocin is the other posterior pituitary hormone, driving milk letdown, uterine contraction, and social bonding. It shares the magnocellular neurosecretory pathway with AVP. Both are nonapeptides differing by two amino acids.

BOARD TRAP

Oxytocin deficiency is clinically silent; do not attribute lactation failure to posterior pituitary unless other features present.

13. DDAVP / desmopressin

WHAT YOU SEE

DDAVP eyedropper, resists vasopressinase

WHAT IT ENCODES

Desmopressin (DDAVP) is a synthetic V2-selective AVP analog with a longer half-life and resistance to vasopressinase. It treats central DI and is used in gestational DI. It lacks significant V1 vasoconstrictor activity.

BOARD TRAP

DDAVP corrects central and gestational DI but NOT nephrogenic DI — failure to respond points to a renal (V2/AQP2) defect.

14. Vasopressinase / gestational DI

WHAT YOU SEE

Vasopressinase placental enzyme, pregnancy 3-4x AVP clearance

WHAT IT ENCODES

Placental vasopressinase degrades native AVP, raising clearance 3-4 fold in pregnancy and causing gestational DI. DDAVP resists this enzyme and is the treatment of choice. It typically resolves postpartum.

BOARD TRAP

Gestational DI is treated with DDAVP (vasopressinase-resistant), not native vasopressin which would be rapidly degraded.

15. Non-osmotic triggers

WHAT YOU SEE

Nausea, hypoglycemia, angiotensin II, smoking listed

WHAT IT ENCODES

Potent non-osmotic AVP stimuli include nausea (very strong), pain, hypoglycemia, angiotensin II, and nicotine. Nausea can raise AVP many-fold and is a common cause of perioperative hyponatremia. These bypass osmoreceptor control.

BOARD TRAP

Postoperative hyponatremia is often nausea/pain-driven ADH plus hypotonic fluids — not necessarily SIADH from a tumor.
